UPPSALA UNIVERSITET
Matematiska institutionen
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Prov i matematik
Geometri och Analys II, 1MA188,
2025-04-14

Time: 08:00 - 13:00. Tools allowed: only pen/pencil and paper.

The exam consists of **8 problems** of **5 points** each for a total of **40 points**. You need 18, 24, 32 points for grade 3, 4, 5, respectively.

Good luck and have fun!

1. (5pts) Let T be the linear transformation in $\mathbb{R}^3$ that reflects with respect to a line ℓ that goes through the origin. We are also given that $T(1,2,3) = (-3,-2,1)$.

(a) Find a directional vector and an equation for the line ℓ.

(b) Find the standard matrix $[T]$ of T.

2. (5pts) Let $0 < x < 1$ be a real constant.

(a) Use Gaussian elimination method to compute the inverse of the matrix

$$A = \begin{pmatrix} x & \sqrt{1-x^2} & 0 \\ \sqrt{1-x^2} & -x & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

if it exists.

(b) Use (a) to compute A^{2025}.

3. Let

$$f(x,y) = \begin{cases} \dfrac{(x-y)^3}{x^2+y^2} & \text{if } (x,y) \neq (0,0), \\ 0 & \text{if } (x,y) = (0,0). \end{cases}$$

(a) (1pt) Find all the points where $f(x,y)$ is continuous. Prove your answer.

(b) (1pt) Compute $f_x(0,0)$ and $f_y(0,0)$.

(c) (1pt) Compute $f_x(x,y)$ for $(x,y) \neq (0,0)$.

(d) (1pt) Check if $f_x(x,y)$ is continuous at $(0,0)$.

(e) (1pt) Compute $D_{\vec{u}}f(0,0)$, the directional derivative of f at $(0,0)$ in the direction of a unit vector $\vec{u}$.
(*Hint: It can be shown that f is not differentiable at $(0,0)$ which is why $D_{\vec{u}}f(0,0)$ is not equal to $\nabla f(0,0) \cdot \vec{u}$ here*)

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4. Consider the curve in $\mathbb{R}^3$ given by the system of equations

$$\begin{cases} x^2 + y^3 + z^4 = 3 \\ xy + xz + yz = 3 \end{cases}$$

(a) (3pts) Show that in some neighbourhood of the point $(x, y, z) = (1, 1, 1)$ we can parametrize this curve using z as the parameter, that is x and y can be viewed as functions of z.

(b) (2pts) Compute $\dfrac{\partial x}{\partial z}$ and $\dfrac{\partial y}{\partial z}$ at the point $(1, 1, 1)$.

5. (5pts) Let $f = f(s, t)$. Solve the partial differential equation

$$s\frac{\partial f}{\partial s} - t\frac{\partial f}{\partial t} = 2s^4 - 2t^4$$

by using the change of variables $x = s^2 - t^2$, $y = 2st$.

6. (5pts) Assume

$$T(x, y) = x^4 + 2y^2 + 1$$

describes the temperature at any point (x, y) on $\mathbb{R}^2$. A bumblebee is traveling along the trajectory $y = x^2 - 3$ and is trying to be cool. At which point of his trajectory should it stop in order to experience the lowest possible temperature along the curve $y = x^2 - 3$? Setup a constrained optimization problem and solve using the Lagrange method of multipliers.

7. (5pts)

(a) Sketch the set D described by the inequalities $x \geq 0$, $y \geq 0$, $y \leq 1$, $y \geq x$, $x^2 + y^2 \geq 1$.

(b) Use polar coordinates to compute the double integral

$$\iint_D \frac{1}{(x^2 + y^2)^{3/2}}\, dxdy$$

(*Hint: express $y = 1$ using the polar coordinates. During your computation you may use that* $\int \frac{dt}{\sin^2 t} = -\frac{\cos t}{\sin t}$)

8. (5pts) Let E be the domain in $\mathbb{R}^3$ bounded by the surfaces $x = 0$, $y = 0$, $z = x + y$, $z = 2 - x - y$. Compute the triple intergral

$$\iiint_E x\, dV$$